STAC PS-3: Exoplanet Hunters

Team Nexus

TABLE I
PROPERTIES OF THE THREE STAC STARS

Star	Mass ($M_\odot$)	Radius ($R_\odot$)	Luminosity ($L_\odot$)	Surface Temp. (K)
STAC 1	1.00	1.00	1.00	5800
STAC 2	0.80	0.80	0.40	5000
STAC 3	0.50	0.50	0.08	3800

***Abstract*—This paper investigates three hypothetical exoplanets orbiting the stars STAC 1, STAC 2, and STAC 3. Using transit photometry, stellar properties, and orbital observations, the planetary radii, orbital distances, habitable-zone reference distances, and equilibrium temperatures are estimated. The results are compared to identify the most promising candidate for further habitability investigations.**

***Index Terms*—Exoplanets, Transit Photometry, Orbital Mechanics, Habitable Zone, Equilibrium Temperature, Planetary Habitability.**

I. PROVIDED DATA

Constants:

- Solar Radius, $R_\odot = 696,000$ km
- Earth Radius, $R_\oplus = 6371$ km
- 1 AU $= 149.6 \times 10^6$ km
- 1 year $= 365.25$ days
- $M_\odot$ = Mass of the Sun
- $L_\odot$ = Luminosity of the Sun
- 1 AU = Distance between Earth and Sun

II. MISSION

In the rest of the document, STAC X (X = 1, 2, 3) refers to the planet orbiting the STAC X star unless mentioned otherwise.

For recorded values that differ from the usual orbital periods, we assume them to be anomalous readings and do not include them in our calculations.

A. Analysis of Transit Depth

The transit depth is calculated using:

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}} \tag{1}$$

where:

- δ = Transit Depth
- F_{normal} = Brightness outside transit
- F_{transit} = Brightness during transit

1) STAC 1:

$$F_{\text{transit}} = 0.997$$
$$F_{\text{normal}} = 1$$
$$\delta_{\text{STAC1}} = \frac{1 - 0.997}{1}$$
$$\boxed{\delta_{\text{STAC1}} = 0.003}$$

The observed dips occur at days 3, 8, 12, 16 and 20. The intervals are approximately 4 days, indicating a periodic transit.

2) STAC 2:

$$F_{\text{transit}} = 0.999$$
$$F_{\text{normal}} = 1$$
$$\delta_{\text{STAC2}} = \frac{1 - 0.999}{1}$$
$$\boxed{\delta_{\text{STAC2}} = 0.001}$$

The observed dips occur at days 20, 100 and 180, indicating a periodic transit every 80 days.

3) STAC 3: The regularly spaced dips occur at days 20, 60, 100 and 140. We use these four observations to calculate the average transit brightness.

$$F_{\text{transit}} = \frac{0.9990 + 0.9992 + 0.9991 + 0.9993}{4}$$
$$F_{\text{transit}} = 0.99915$$
$$F_{\text{normal}} = 1$$
$$\delta_{\text{STAC3}} = \frac{1 - 0.99915}{1}$$
$$\boxed{\delta_{\text{STAC3}} = 0.00085}$$

B. Determination of Planet Size

The planetary radius is estimated using:

$$\delta \approx \left(\frac{R_p}{R_s}\right)^2 \tag{2}$$

Rearranging:

$$R_p = R_s\sqrt{\delta} \tag{3}$$

where:

- R_p = Radius of the Planet
- R_s = Radius of the Star
- δ = Transit Depth
- For Reference (Radii):
  - Jupiter: 69,911 km
  - Neptune: 24,764 km

- Super-Earth: approximately 8,000 km to 12,750 km
- Earth: 6,371 km

1) STAC 1:

$$R_p = 1 \times 696000 \times \sqrt{0.003}$$
$$R_p \approx 38121.49 \text{ km}$$
$$R_p = \frac{38121.49}{6371}$$
$$\boxed{R_p \approx 5.98 R_\oplus}$$

Category = Large Neptune / Sub-Saturn Sized.

2) STAC 2:

$$R_p = 0.8 \times 696000 \times \sqrt{0.001}$$
$$R_p \approx 17607.56 \text{ km}$$
$$R_p = \frac{17607.56}{6371}$$
$$\boxed{R_p \approx 2.76 R_\oplus}$$

Category = Sub-Neptune Sized.

3) STAC 3:

$$R_p = 0.5 \times 696000 \times \sqrt{0.00085}$$
$$R_p \approx 10145.86 \text{ km}$$
$$R_p = \frac{10145.86}{6371}$$
$$\boxed{R_p \approx 1.59 R_\oplus}$$

Category = Super-Earth Sized.

C. Determination of Planetary Orbits

Kepler's Third Law is used:

$$P^2 = \frac{a^3}{M} \tag{4}$$

Rearranging:

$$a = \sqrt[3]{MP^2} \tag{5}$$

where:

- a = Orbital Distance (AU)
- P = Orbital Period (Years)
- M = Mass of the Star (Solar Masses)
- 1 year = 365.25 days
- We assume observations that significantly deviate from the periodic pattern to be anomalous and omit them from orbital-period calculations.

1) STAC 1:

$$P_{\text{orbital}} = 4 \text{ days}$$
$$a = \left(\left(\frac{4}{365.25} \right)^2 \times 1.00 \right)^{1/3}$$
$$\boxed{a \approx 0.04932 \text{ AU}}$$
$$a = 0.0493151 \times 149.6 \times 10^6$$
$$\boxed{a \approx 7,377,538 \text{ km}}$$

2) STAC 2:

$$P_{\text{orbital}} = 80 \text{ days}$$
$$a = \left(\left(\frac{80}{365.25} \right)^2 \times 0.80 \right)^{1/3}$$
$$\boxed{a \approx 0.33731 \text{ AU}}$$
$$a = 0.3373105 \times 149.6 \times 10^6$$
$$\boxed{a \approx 50,461,648 \text{ km}}$$

3) STAC 3:

$$P_{\text{orbital}} = 40 \text{ days}$$
$$a = \left(\left(\frac{40}{365.25} \right)^2 \times 0.50 \right)^{1/3}$$
$$\boxed{a \approx 0.18168 \text{ AU}}$$
$$a = 0.1816784 \times 149.6 \times 10^6$$
$$\boxed{a \approx 27,179,081 \text{ km}}$$

All three planets orbit significantly closer to their host stars than Earth does to the Sun.

D. Determination of Most Promising Planet

The approximate habitable-zone reference distance is:

$$d_{\text{HZ}} \approx \sqrt{\frac{L}{L_\odot}} \text{ AU} \tag{6}$$

where:

- d_{HZ} = Approximate reference distance of the habitable zone
- L = Luminosity of the Star
- $L_\odot$ = Luminosity of the Sun

The difference between the planet's orbital distance and the estimated habitable-zone reference distance is:

$$\Delta d = |d_{\text{HZ}} - a| \tag{7}$$

1) STAC 1:

$$d_{\text{HZ}} = \sqrt{1.00}$$
$$d_{\text{HZ}} = 1.00 \text{ AU}$$
$$\Delta d = |1.00 - 0.04932|$$
$$\boxed{\Delta d \approx 0.95068 \text{ AU}}$$

2) STAC 2:

$$d_{\text{HZ}} = \sqrt{0.40}$$
$$d_{\text{HZ}} \approx 0.63246 \text{ AU}$$
$$\Delta d = |0.63246 - 0.33731|$$
$$\boxed{\Delta d \approx 0.29515 \text{ AU}}$$

3) STAC 3:

$$d_{\text{HZ}} = \sqrt{0.08}$$
$$d_{\text{HZ}} \approx 0.28284 \text{ AU}$$
$$\Delta d = |0.28284 - 0.18168|$$
$$\boxed{\Delta d \approx 0.10116 \text{ AU}}$$

STAC 3 is the closest to its estimated habitable-zone reference distance. Its Super-Earth size also makes it an interesting candidate for further investigation.

However, this simplified formula does not calculate the actual inner and outer boundaries of the habitable zone. Therefore, it cannot independently establish whether a planet is inside the habitable zone.

E. Determination of Planet Equilibrium Temperature

The planetary equilibrium temperature is calculated using:

$$T_{\text{eq}} = T_s \times \left(\frac{R_s}{2d}\right)^{1/2} \times (1 - A)^{1/4} \tag{8}$$

where:

- T_{eq} = Planetary Equilibrium Temperature
- T_s = Surface Temperature of the Star
- R_s = Radius of the Star
- d = Orbital Distance (in km)
- A = Bond Albedo (Reflectivity of the Planet)
- Since no albedo values are provided, we assume an Earth-like Bond Albedo of 0.3.

1) STAC 1:

$$T_{\text{eq}} = 5800 \times \left(\frac{696000}{2 \times 7377538}\right)^{1/2} \times (1 - 0.3)^{1/4}$$
$$\boxed{T_{\text{eq}} \approx 1152 \text{ K}}$$
$$T_{\text{eq}} = 1152.22 - 273.15$$
$$\boxed{T_{\text{eq}} \approx 879^\circ\text{C}}$$

2) STAC 2:

$$T_{\text{eq}} = 5000 \times \left(\frac{556800}{2 \times 50461648}\right)^{1/2} \times (1 - 0.3)^{1/4}$$
$$\boxed{T_{\text{eq}} \approx 340 \text{ K}}$$
$$T_{\text{eq}} = 339.70 - 273.15$$
$$\boxed{T_{\text{eq}} \approx 67^\circ\text{C}}$$

TABLE II
COMPARISON OF THE THREE PLANETS

Planet	Distance to HZ Reference	Equilibrium Temp.	Temp. (°C)
STAC 1	0.95068 AU	1152 K	879°C
STAC 2	0.29515 AU	340 K	67°C
STAC 3	0.10116 AU	278 K	5°C

3) STAC 3:

$$T_{\text{eq}} = 3800 \times \left(\frac{348000}{2 \times 27179081}\right)^{1/2} \times (1 - 0.3)^{1/4}$$
$$\boxed{T_{\text{eq}} \approx 278 \text{ K}}$$
$$T_{\text{eq}} = 278.11 - 273.15$$
$$\boxed{T_{\text{eq}} \approx 5^\circ\text{C}}$$

III. FINAL CONCLUSION

Based on the calculated equilibrium temperatures:

- **STAC 1:** Extremely hot ($\sim 879^\circ$C), making it a very unlikely candidate for liquid surface water.
- **STAC 2:** Relatively warm ($\sim 67^\circ$C). Its habitability would depend heavily on atmospheric conditions and heat distribution.
- **STAC 3:** Relatively cool ($\sim 5^\circ$C), making it the most promising candidate. A suitable atmosphere and greenhouse effect could help maintain conditions favourable for liquid water.

Since Princess Leia and the Rebels are humans (excluding the droids, who can survive in a wider range of environments), the fleet should hyperdrive towards **STAC 3**. Its Super-Earth-sized planet and relatively favourable estimated equilibrium temperature make it the strongest candidate for further habitability investigations.

An additional observation that could help determine its suitability is atmospheric spectroscopy. This would help identify gases such as water vapour, carbon dioxide, oxygen and methane, providing information about its atmosphere and possible habitability.

The planets of STAC 1 and STAC have an estimated radius above 1.75 R_E (Earth's Radius). According to modern astrophysics, planets this large are more likely to have a substantially dense and large gaseous atmosphere rather than being entirely rocky. This is because sufficiently massive planetary cores can attract and accumulate large amounts of gas during their formation. As more gas accumulates, the planet's mass and gravitational attraction increase, potentially allowing it to capture even more gas. STAC-1, with an estimated radius of approximately 5.98 R_E, is likely to be a gas giant, while STAC-2, at approximately 2.76 R_E, may be classified as a sub-Neptune or gas-rich planet. However, their exact internal compositions cannot be determined from their radii alone. Even if their gaseous envelopes were stripped away by intense stellar radiation, exposing their rocky interiors, their surface conditions would still depend on factors such as temperature

and atmospheric composition. Therefore, neither planet can be assumed to provide suitable living conditions for the Rebel crew. Even the planet of the star, STAC - 3, is likely to have a high atmospheric pressure and would require Princess Leia and the rebel crew to check for that before they would be able to settle on it.

Note: Equilibrium temperatures are approximate values and do not necessarily represent actual surface temperatures. Atmospheric composition, greenhouse effects, clouds and planetary rotation can significantly affect actual real-world conditions.